

# Response to Reviewers

## Editor comments

After a round of revisions, both reviewers agree that the manuscript has greatly improved, but there are still some issues that need to be addressed by the authors.

-Rongji Mu

## Response to the Editor

We would like to thank the editor and the two reviewers for their time and careful reading of our manuscript. We believe that their valuable feedback and the revisions we made in response have substantially improved our work. We respond to each point with the modified text quoted below our respective reply.

## Reviewer 1

The authors have adequately addressed my previous comments, and the manuscript has been substantially improved. I only have one minor comment.

### Minor comments

#### Reviewer 1, comment 1

The statement in the Abstract that “frequentist meta-analytic approaches ... often fall short in this regard” seems somewhat unbalanced. I suggest revising this statement to present a more objective comparison between frequentist and Bayesian approaches.

### Response

We thank the reviewer for this comment and agree that our original statement in the abstract could be improved. We have revised the paragraph.

### Revised: Abstract - Background

*“Uncertainty quantification is central to evidence synthesis for risk assessment and policy-making. Frequentist meta-analysis - currently predominant in epidemiology - quantifies uncertainty through confidence intervals, while Bayesian meta-analysis provides posterior distributions. We performed frequentist and Bayesian meta-analyses to compare uncertainty quantification and applicability for risk assessment using lead exposure and children’s IQ as a case study.”*

## Reviewer 2

### Major comments

#### Reviewer 2, comment 1

Please address the omitted request for frequentist prediction intervals and Bayesian posterior

predictive intervals, particularly given the very high heterogeneity and the manuscript's focus on uncertainty quantification.

### **Response**

We have now added frequentist and Bayesian predictive intervals to the manuscript and revised the methods, results, and discussion accordingly. We thank the reviewer for suggesting this addition and agree that it is an important addition to our manuscript.

### **Revised: Methods - Bayesian Meta-Analysis**

*"To quantify uncertainty in the pooled mean and heterogeneity jointly, we constructed the posterior predictive distribution of hypothetical future studies by sampling from  $\mathcal{N}(\mu, \tau)$ . Additionally, a sensitivity analysis with alternative priors was conducted, as described below."*

### **Revised: Methods - Frequentist Meta-Analysis**

*"In the frequentist approach, we constructed the predictive distribution of hypothetical future studies via parametric simulation. To enable parametric simulation from point estimates, we assumed a normal distribution for  $\mu$  and a positive half-normal distribution for  $\tau$ , sampled from these distributions, and then used these samples to draw from the predictive distribution  $\mathcal{N}(\mu, \tau)$ . The number of draws in the parametric simulation matched the number of MCMC samples from the Bayesian approach."*

### **Revised: Results**

*"The Bayesian random-effects meta-analysis yielded a posterior mean pooled estimate of  $-2.28$  (95% credible interval (CrI):  $-3.54$  to  $-1.04$ ) IQ points per log-unit increase in BLL ( $\mu\text{g/dL}$ ) (see Fig. 2) and a 95% predictive interval from  $-6.28$  to  $1.69$ . The frequentist approach produced a comparable pooled estimate of  $-2.45$  (95% confidence interval (CI):  $-3.82$  to  $-1.08$ ) and a comparable 95% predictive interval from  $-6.53$  to  $1.53$ ."*

### **Revised: Discussion**

*"The option to create predictive distributions that can inform EBD studies is, however, not exclusive to the Bayesian approach. In the frequentist framework, they can be obtained using, for example, parametric simulation, nonparametric bootstrap, or the delta method. Each method comes with its own assumptions that may impact the results, requiring the researcher to carefully weigh these decisions. In this study, the small sample size led to convergence issues when attempting to use nonparametric bootstrap. We therefore estimated the predictive interval using parametric simulation because it was feasible in our setting and procedurally similar to the Bayesian approach, making it more comparable. We assumed normal and truncated normal distributions for  $\mu$  and  $\tau$ , respectively, to match the Bayesian setup but we could have theoretically assumed any distribution for  $\tau$  whose support is limited to positive real values. If we had instead chosen a log-normal distribution for  $\tau$ , the predictive interval would have reached from  $-6.74$  to  $1.74$ . By assuming distributions for  $\mu$  and  $\tau$  after conditioning on the data, the data*

*cannot inform the assumed distributions during parameter estimation. The Bayesian approach differs in that prior distributions will, given sufficient data, converge to the true distribution so long as that prior places non-zero probability on the true values [63].”*

*“We also demonstrate how both approaches quantify uncertainty in predictive quantities, exemplified by calculating the distribution of hypothetical new studies.”*

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#### **Reviewer 2, comment 2**

Please address the previously raised systematic-review concern, especially the literature search ending in July 2023 and the use of a single human reviewer for full-text screening/RoB assessment. (One of prior comments: The literature search ended in July 2023, although the manuscript includes references from 2025 and appears to have been prepared later. The search should be updated.)

#### **Response**

We thank the reviewer for this comment and agree that the search should be updated, which we did by conducting a rapid search in PubMed for the period from July 2023 to 2 September 2026, using the original search terms. All titles were first screened by one reviewer and, afterwards, by the LLM Claude Opus 5. If the two judgements diverged, the record was revisited by the researcher, who made the final decision. Our search returned 173 records, none of which met the PECOTS and eligibility criteria. No study was therefore added to the meta-analysis and the pooled estimates are unchanged. We have added the update in the manuscript and additional file 3. Because the update was narrower in scope than the original search, covering one database and relying on a single reviewer, we cannot fully rule out that a comprehensive update of the literature search would have identified further studies, but we consider it unlikely that such a comprehensive update would materially change our results. We have also made the limitation of only one human reviewer for the full-text screening and the RoB assessment explicit in the limitations paragraph. This comment also helped us realise that a core objective of our study, which is to compare uncertainty quantification of frequentist and Bayesian meta-analyses, was not stated clearly enough. We have therefore revised the aims statement at the end of the Background section, as well as the opening sentence of the Discussion, and the Conclusions section. We would like to thank the reviewer once again for their comment and believe that the changes made have greatly improved the manuscript.

#### **Revised: Background**

*“The aim of this study is to compare uncertainty quantification of frequentist and Bayesian meta-analyses. We apply both meta-analytic approaches to the association between BLL and IQ scores in children using the results of a systematic review on epidemiological studies published between January 2005 and July 2023, thereby also providing an updated pooled effect estimate of this association.”*

**Revised: Methods - Literature Review**

*“We conducted a rapid update search in PubMed in September 2026, using the same queries as the original search and restricted to records published after July 2023. Titles were first screened by one reviewer and additionally by the large language model (LLM) Claude Opus 5 (Anthropic); when the two judgements deviated, the reviewer revisited the record and made the final decision.”*

**Revised: Results - Literature Review**

*“The rapid update search in PubMed yielded 173 records published after July 2023, none of which met the eligibility criteria. 44 and 40 records were excluded because the exposure or outcome definition did not match, respectively; 30 records were excluded because the record was a review rather than a primary study and 59 records were excluded for other reasons, including animal studies and studies on the mechanisms of lead neurotoxicity. Therefore, no new studies were added to the meta-analysis. Screening decisions for the update are documented in additional file 3.”*

**Revised: Discussion**

*“In this study, we compare uncertainty quantification of a frequentist and a Bayesian meta-analysis, which is less used in the field of environmental epidemiology to date, using lead exposure and children’s IQ as an applied example.”*

*“Our original literature search was limited to January 2005 to July 2023 and was updated in September 2026 through a rapid search in PubMed, which identified no additional eligible studies. Although this update was narrower in scope than the original search, it is unlikely that a full update covering both databases would materially change the pooled estimate. Due to resource constraints, full-text screening and RoB assessment could not be carried out by multiple independent persons.”*

*“However, these limitations did not affect the main objective of this study, which was to compare uncertainty quantification in frequentist and Bayesian meta analysis in an applied setting.”*

**Revised: Conclusion**

*“This study provides an applied methodological comparison of Bayesian and frequentist meta-analysis and an evidence synthesis of epidemiological studies examining the association between lead exposure and children’s IQ.”*

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**Reviewer 2, comment 3**

Please revise the remaining statistical overstatements regarding shrinkage as “giving less weight,” frequentist weighting being “not configurable,” and high heterogeneity being attributable simply to a small number of studies.

**Response**

We thank reviewer 2 for pointing out these remaining overstatements. We have described the

mechanism of shrinkage more precisely, added detail on how frequentist weighting may be configured, and deleted the earlier incorrect statement that high heterogeneity is a function of the number of studies.

### **Revised: Discussion**

*“The two frameworks may differ in how they treat studies reporting substantial uncertainty depending on prior configuration and sample size. Bayesian meta-analysis can be set up such that it is less sensitive to uncertain studies via the prior distribution for the heterogeneity: the more concentrated this prior is towards zero the stronger the shrinkage, pulling uncertain studies more towards the pooled estimate than certain studies (Table 4). The frequentist meta-analysis can configure the weighting of individual studies by changing the estimation methods for  $\tau^2$ . At the time of writing, the `metafor` package included a list of references pointing to publications for each of these estimation methods. However, a vignette to illustrate the use of these methods to make them more accessible to applied researchers was not available [60].”*